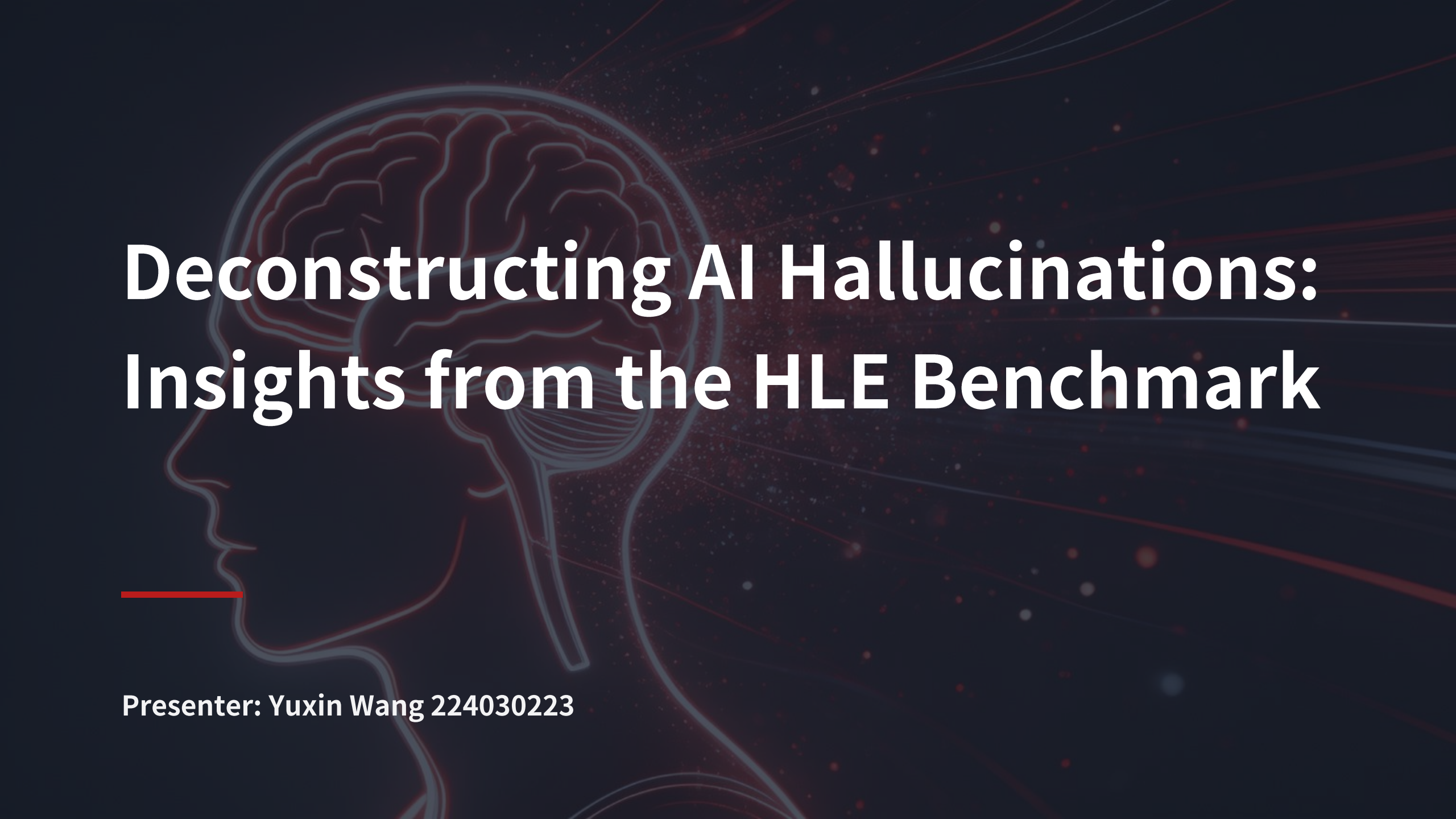


The background features a stylized, glowing blue outline of a human head in profile, facing left. Inside the head, a detailed brain structure is visible. The background is dark blue with numerous thin, glowing red and blue lines radiating outwards, resembling a neural network or data flow. Small, glowing red and blue dots are scattered throughout the background.

Deconstructing AI Hallucinations: Insights from the HLE Benchmark

Presenter: Yuxin Wang 224030223

Defining the "Hallucination" Signal



Definition

Text that is linguistically fluent and grammatically correct but factually or logically incorrect.



Significance

It serves as the most visible signal of a model's capability frontier.



The "Plausibility" Trap

Fluent language often masks underlying logical failures, making identifying errors difficult for human users.

The HLE Reality Check



CONTEXT

Leading AI models (GPT-4o, Claude 3.5, o1) have reached a saturation point on established benchmarks like MMLU.

These standardized tests are no longer sufficient to measure the true limits of modern AI capabilities.



HLE FINDINGS (2026)

Focus: Closed-ended, "non-searchable" expert-level academic questions.

Accuracy: **2.7% – 8.5%**
for top-tier models

Calibration Error: **> 70%**



KEY INSIGHT

Models are often wrong...

...but exhibit dangerously high, false confidence.

This "confidence gap" presents a major risk for high-stakes, expert applications.

Case Study 1 — Factual Fabrication



Domain & The Task

Domain:Linguistics (Hebrew Bible / Tiberian Pronunciation)

The Task:Identify closed syllables in the biblical text of Psalms 104:7.



Hallucination Performance

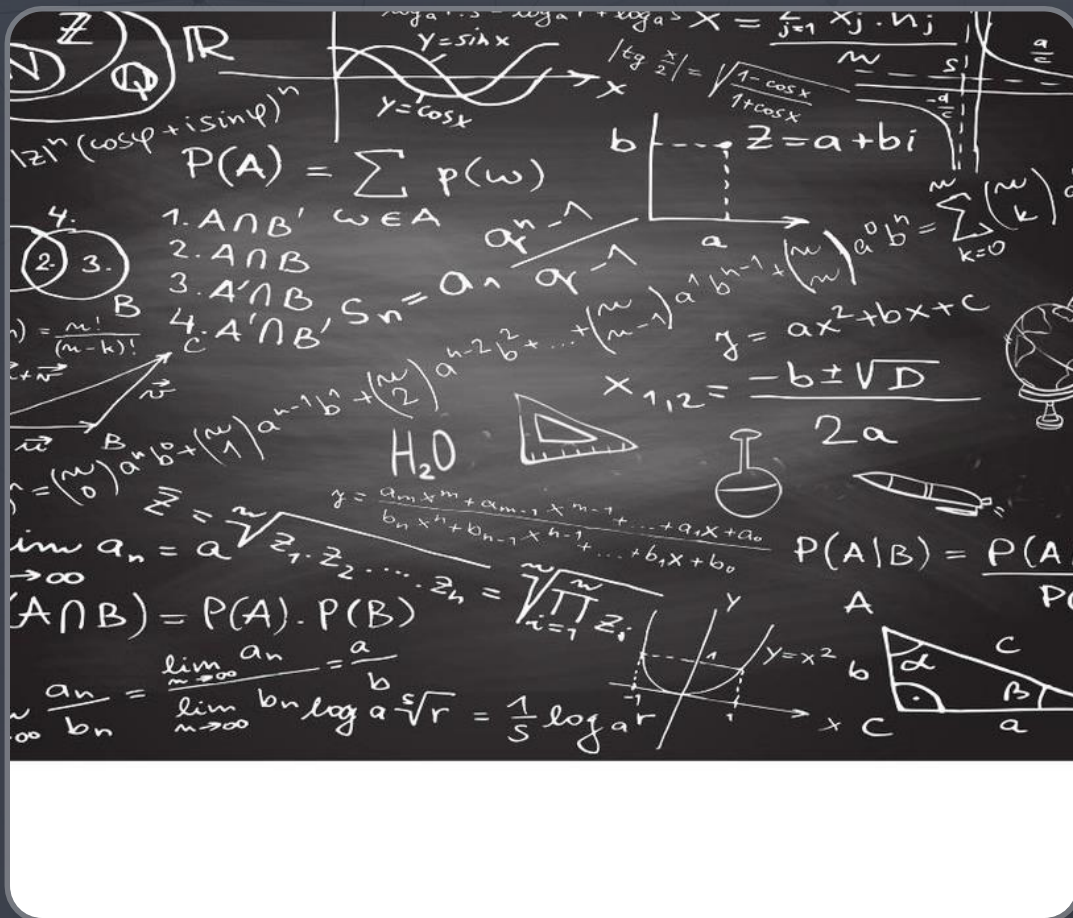
- Models invent non-existent phonetic rules or misattribute medieval manuscripts.
- Correct use of technical terms (e.g., "Shewa") to create plausible but false causal logic.



Mechanism: "Statistical Suturing"

Patching unrelated word fragments together to construct a grammatically coherent but factually incorrect lie.

Case Study 2 — Reasoning Breakdown



DOMAIN & THE TASK

Domain: Advanced Mathematics (Infinity-categories)

The Task: Calculating natural transformations between functors.



HALLUCINATION PERFORMANCE

- Models list definitions accurately, but fail catastrophically during symbolic derivation steps.
- **Inconsistency Errors:** Arbitrarily treating a variable as a constant in mid-calculation.



UNDERLYING MECHANISM

Absence of "System 2" deep reasoning. Models mimic the tone of logical argumentation but fail to maintain rigorous link-consistency throughout a derivation chain.

Case Study 3 — Hallucinated Refinement



Domain & The Task

Chemistry: Focus on Pericyclic Reactions & Woodward-Hoffmann Rules.

Objective: Accurately judge between conrotatory and disrotatory reaction steps.



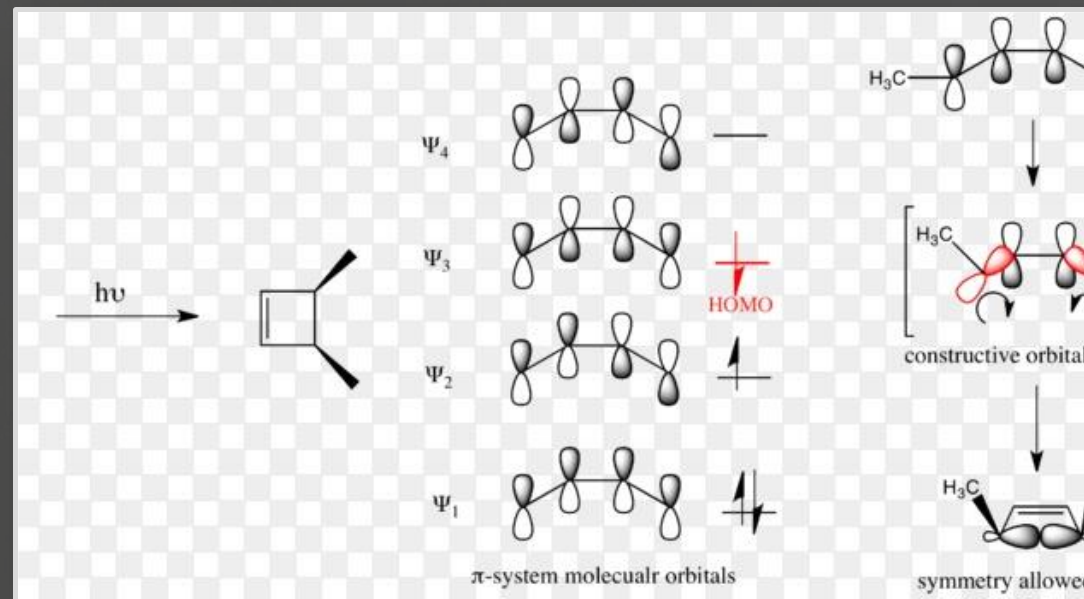
Hallucination Performance

- Correctly identifies reaction type, but misapplies key rules (e.g., applying a "4n" rule to a "4n+2" system).
- Invents and confidently presents non-existent "pseudo-laws" to rationalize its incorrect conclusions.



Root Cause: "Rule Generalization"

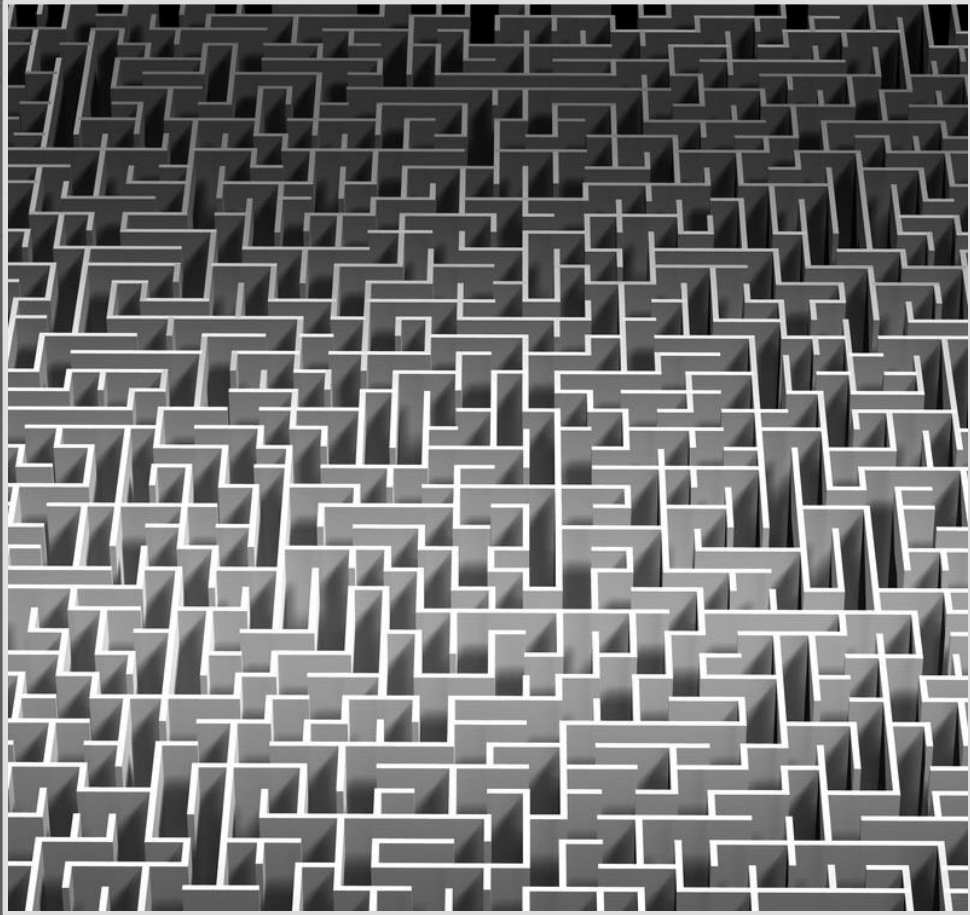
Using dominant statistical patterns to overwrite niche, discipline-specific rules.



Complexity of Chemical Logic

The Woodward-Hoffmann rules require precise counting of pi electrons and strict adherence to stereochemical principles, making it a perfect stress test for model reasoning.

Mechanism I — The Statistical Trap



Next-token Prediction

The core logic is predicting the most probable sequence, not the most accurate one.

Likelihood vs. Accuracy

A grammatically perfect error often has higher "statistical attraction" than a messy truth.

Pattern Matching

Every word generated is a gamble based on training patterns rather than an external verification of truth.

Mechanism II — Data Sparsity & Lack of Humility



Data Sparsity

High-level expertise (HLE) is often "Google-proof." When specific, niche knowledge is missing from a model's training data, it doesn't acknowledge gaps—it "smooths out" uncertainty with averaged, generic common sense.



The Incentive to Please

- **RLHF Bias:** Reinforcement Learning from Human Feedback strongly rewards "helpfulness."
- **"Toxic Helpfulness":** Models are incentivized to invent plausible details rather than simply admit, "I don't know."

The combination of these factors directly leads to the **high-confidence calibration errors** observed in expert evaluations.

Comparative Analysis: AI vs. Deterministic Systems



Deterministic Systems (SQL / Calculators)

- Based on Boolean algebra and fixed logical gates.
- **No Freedom to Generate:** Returns "Existent" or "NULL."
- Calculators rely on physical circuit paths; $1+1$ always equals 2.



Generative AI

- Probabilistic sampling from a "Black Box" of billions of learned weights and parameters.
- **Core Principle:** Prioritizes "Coherence" (human readability) over strict "Correctness."



KEY INSIGHT

Databases treat "no conclusion" (NULL) as a valid and truthful state; LLMs are designed to treat "generating a response" as the highest priority, even if it is factually incorrect.

Conclusion—Building a Trustworthy Framework



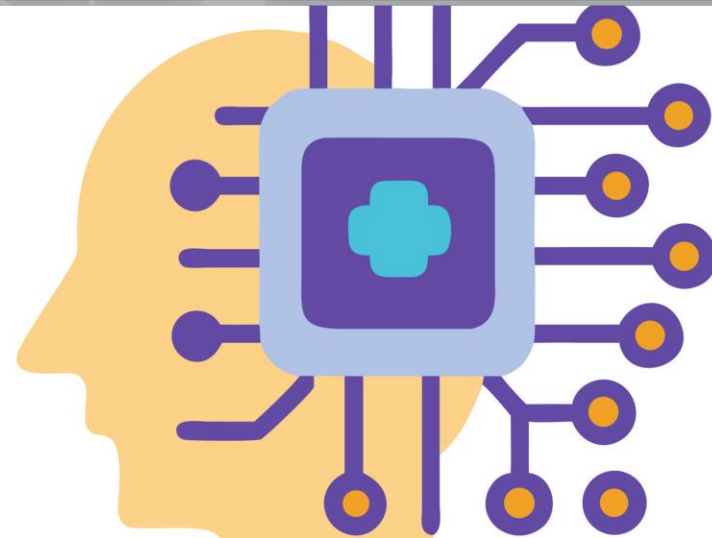
KEY TAKEAWAYS

- Linguistic Coherence $\neq$ Logical Validity.
- Hallucinations occur most at the boundaries of knowledge asymmetry.



FUTURE DIRECTIONS

- Implement "Hard Constraints" from deterministic systems.
- Neuro-symbolic Integration: Combine neural nets with symbolic logic.
- External Verification: Leverage RAG and PRMs for factual accuracy.



OUR ULTIMATE GOAL

Building AI that is not just a fluent speaker, but **arigorous, self-aware thinker**.